

Unit 4: Elliptic Curves

Outline

1. Introduction to Elliptic Curves
2. Operations on elliptic curves
3. Elliptic curves in finite bodies
4. Cryptography with elliptic curves

Introduction to Elliptic Curves

Introduction

- Technological advances force us to use larger RSA keys. Nowadays we talk about 4096 bits being needed to be secure
 - RSA is a safe and tested standard, but a bit computationally "heavy"
- Elliptic curves offer equivalent security with smaller keys, but have been tested for less time
- Because of this, elliptic curve cryptography is mainly used in situations that need efficiency
 - Asymmetric cryptography is mostly used for key and digital signature exchanges, but some websites have a lot of traffic
 - They need to establish a lot of shared keys!
 - Because of this companies like Google use Elliptic Curves for efficiency, while banks stick with RSA for security

Elliptic Curves

- An elliptic curve is not an ellipse
- It is a curve $E(a,b)$ that has an equation of degree 3

$$y^2 = x^3 + ax + b$$

- To draw it, we calculate

$$y = \sqrt{x^3 + ax + b}$$

- It has symmetry with respect to the $y=0$ axis
- In addition to the points on the curve, we also have the point "to infinity" or O
- We say an elliptic curve is an abelian group with respect to the sum

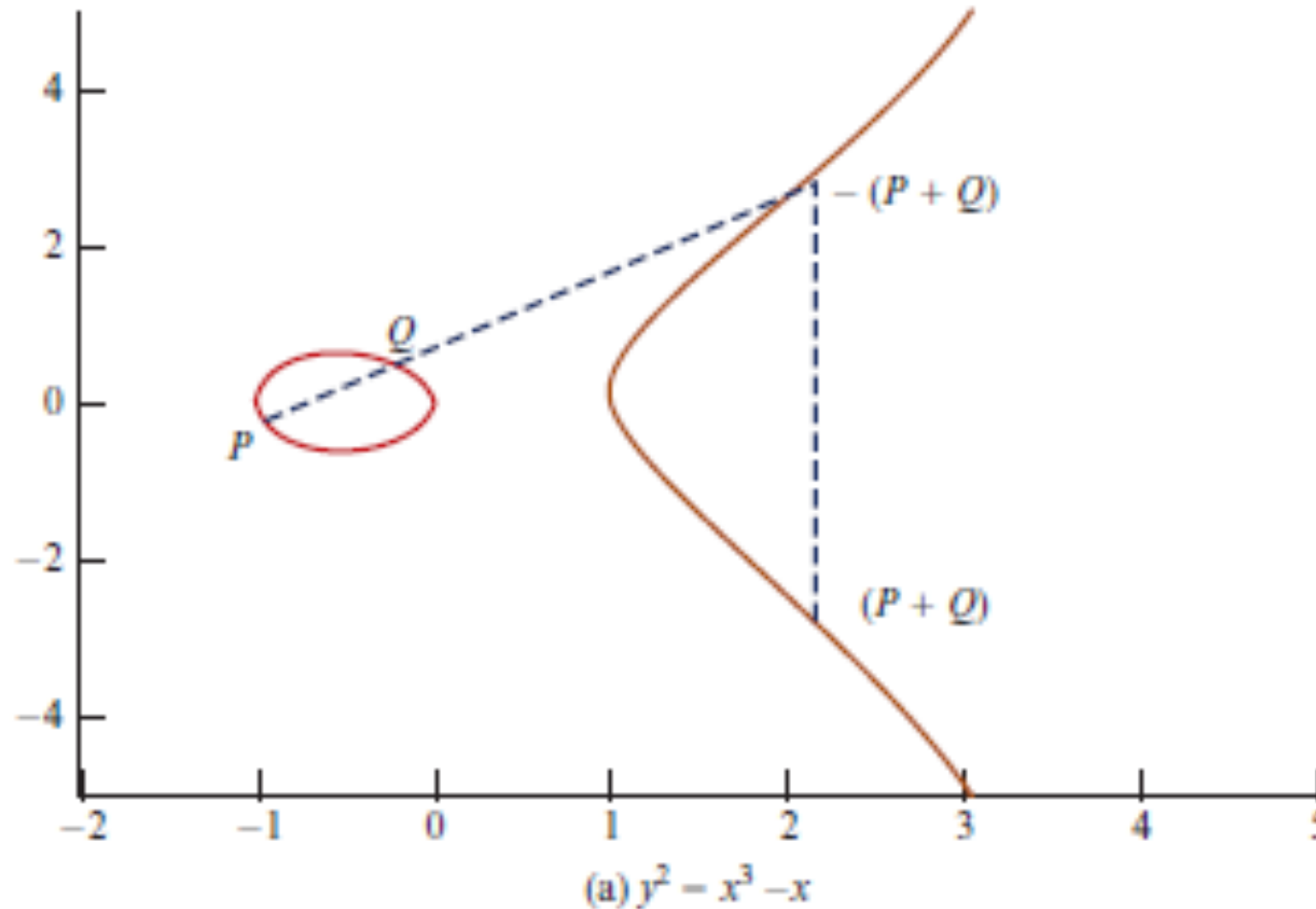
Groups

- A group $\mathbb{G}$ is a set that has an operation (in our case the sum) with the following properties
 - Closure: If a and b belong to $\mathbb{G}$ then $a + b$ also belong to $\mathbb{G}$
 - Associative: $(a + b) + c = a + (b + c)$
 - There exists an element (identity, 0) such that: $a + 0 = 0 + a = a$
 - Each element in the group has an inverse such that $a + b = 0$
- Also, if the group satisfies the commutative property, it is an abelian group
- Examples for the addition operation
 - Integers $\mathbb{Z}$ are an abelian group
 - Natural numbers $\mathbb{N}$ aren't. Why?

Addition on elliptic curves

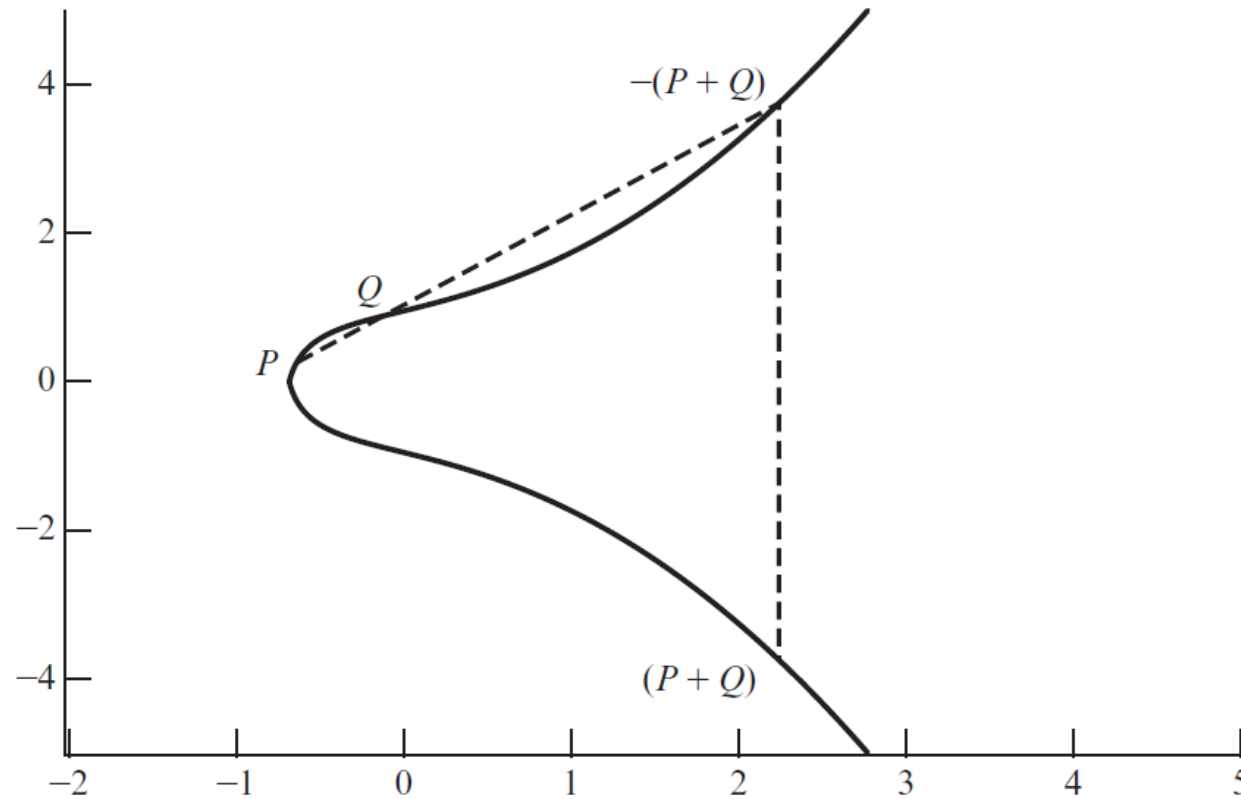
- A point on an elliptic curve is defined by its x and y coordinates: (x,y)
- On an elliptic curve we don't add the coordinates directly
 - It doesn't make sense
- Additions on elliptic curves
 - The sum of two points P and Q is defined as follows
 - $P+Q+R = 0 \rightarrow P+Q = -R$
 - We draw a line from P to Q until it intersects with the curve at R again
 - We take the symmetrical with R which will be -R

Example: Curve $E(-1,0)$: $y^2 = x^3 \pm 1x$



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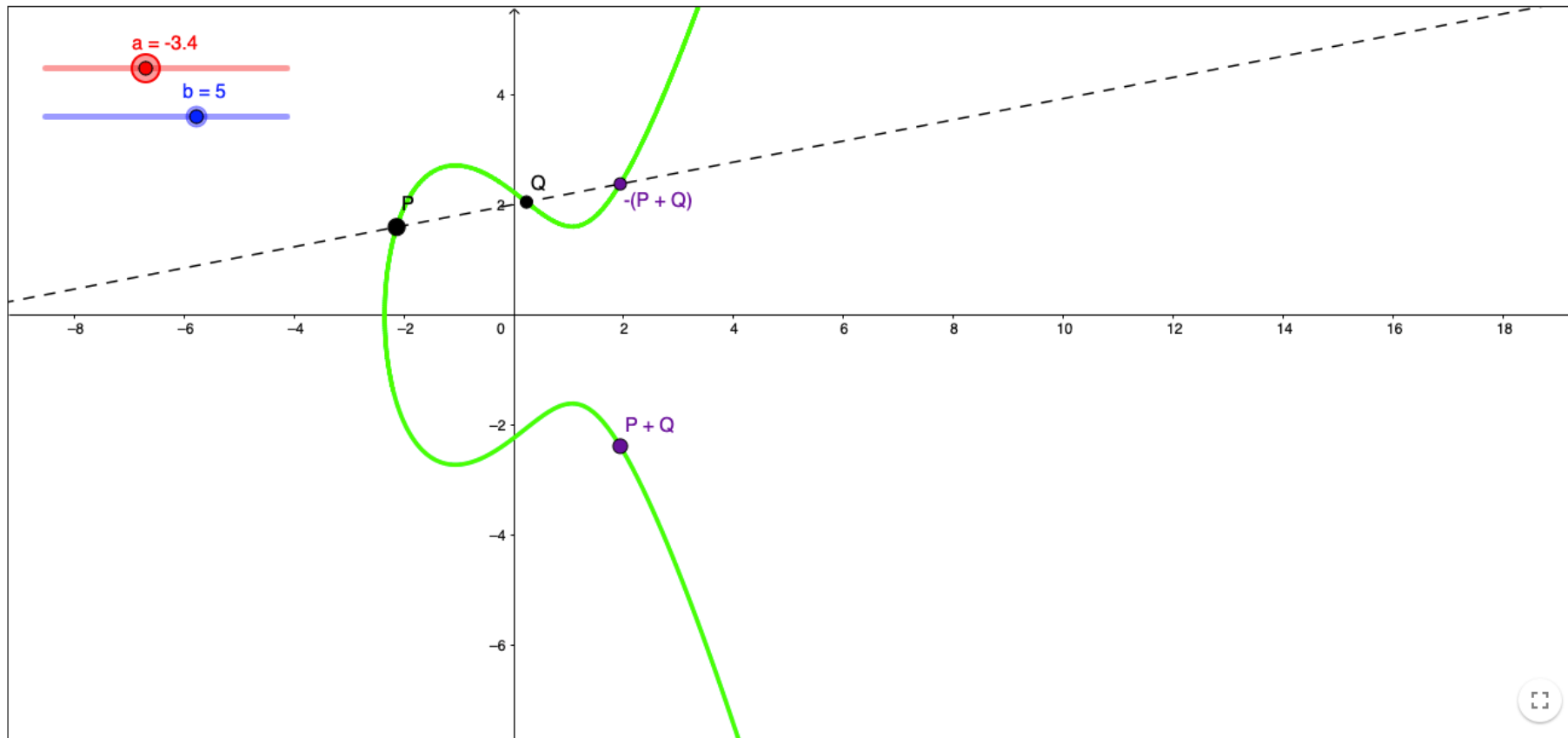
Example: curve $E(1,1) = y^2 = x^3 + x + 1$



(b) $y^2 = x^3 + x + 1$

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Other examples



Other Arithmetic Operations on Elliptic Curves

- Adding two points is relatively easy
- Two inverse points add up to 0
 - $R - R = 0$
- But what if I want to add the same point?
 - $P+P+R = 0 \rightarrow 2P = -R$
 - To add the same point we use the tangent to the curve at that point
 - Intuition: The tangent is the boundary of the lines passing through two points infinitely close to P
- In this way we can define multiplication in a similar way
 - $kP = P+P+\dots+P$ k times.

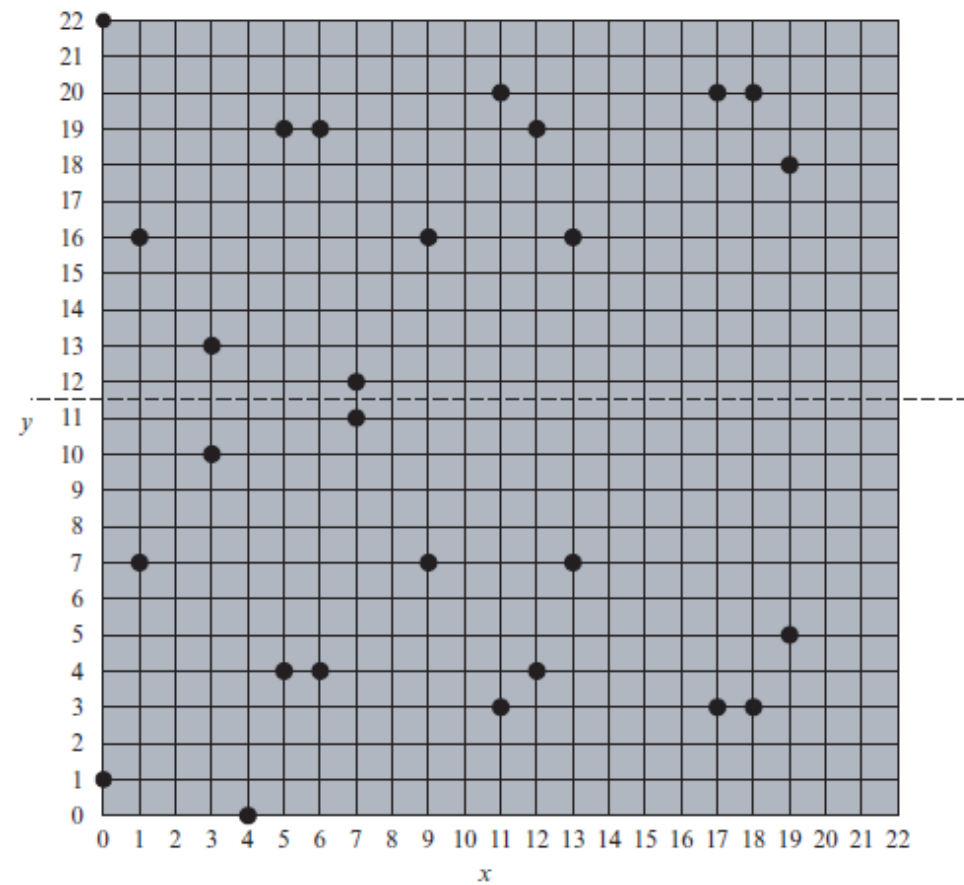
Elliptic curves in finite bodies (mod p)

- We can use elliptic curves on finite bodies (mod p)
- If we take only coefficients and variables in a prime size field p, we have an elliptic curve $E_p(a, b)$
$$y^2 = x^3 + ax + b \text{ mod } p$$
- It is still a group but
 - His drawing is not that easy
 - Operations are described algebraically as calculations mod p

Example: $E_{23}(1,1) \ y^2 = x^3 + x + 1 \mod 23$

(0, 1)	(6, 4)	(12, 19)
(0, 22)	(6, 19)	(13, 7)
(1, 7)	(7, 11)	(13, 16)
(1, 16)	(7, 12)	(17, 3)
(3, 10)	(9, 7)	(17, 20)
(3, 13)	(9, 16)	(18, 3)
(4, 0)	(11, 3)	(18, 20)
(5, 4)	(11, 20)	(19, 5)
(5, 19)	(12, 4)	(19, 18)

Example: $E_{23}(1,1)$ $y^2 = x^3 + x + 1 \mod 23$



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Otra [web](#)

Exercise 1

- Let's use the following online tool:
<https://andrea.corbellini.name/ecc/interactive/modk-add.html>
- Using $E_{23}(1,4)$. Calculate:
 - $(10,5) + (4,16) =$
 - $(10,5) + (10, 18) =$
 - $(7,3) + (7,3) =$
 - $(22,18) + (7,3) =$
 - $(7,3) + (5,2) =$

Exercise 1

- Let's use the following online tool:
<https://andrea.corbellini.name/ecc/interactive/modk-add.html>
- Using $E_{23}(1,4)$. Calculate:
 - $(10,5) + (4,16) = (13, 12)$
 - $(10,5) + (10, 18) = 0$ (Son inversos)
 - $(7,3) + (7,3) = (22,18)$
 - $(22,18) + (7,3) = (18,9)$
 - $(7,3) + (5,2) = \text{¡Error, (5,2) no pertenece a la curva!}$

Exersize 2

- Now using products:
<https://andrea.corbellini.name/ecc/interactive/modk-mul.html>
- Using $E_{23}(1,4)$. Calculate:
 - $10 \times (11,9) =$
 - $66 \times (8,8) =$
 - $29 \times (1,11) =$

Exercise 2

- Now using products:
<https://andrea.corbellini.name/ecc/interactive/modk-mul.html>
- Using $E_{23}(1,4)$. Calculate:
- $10 \times (11,9) = (0,2)$
- $66 \times (8,8) = (17,9)$
- $29 \times (1,11) = 0$ (¿por qué?)

Cryptography with Elliptic Curves

Cryptography with Elliptic Curves

- The addition in elliptic curves in finite fields equivalent to modular multiplication
- In this way, the multiplication by a constant is equivalent to modular exponentiation
- The trap function in this case is the product $Q = kP$
 - It is easy to get Q knowing k y P
 - It is difficult to find k knowing Q y P
- This is the “discrete logarithm problem” but for elliptic curves

Example: Diffie-Hellman

Global Public Elements

$E_q(a, b)$	elliptic curve with parameters a, b , and q , where q is a prime or an integer of the form 2^m
G	point on elliptic curve whose order is large value n

User A Key Generation

Select private n_A	$n_A < n$
Calculate public P_A	$P_A = n_A \times G$

User B Key Generation

Select private n_B	$n_B < n$
Calculate public P_B	$P_B = n_B \times G$

Calculation of Secret Key by User A

$$K = n_A \times P_B$$

Calculation of Secret Key by User B

$$K = n_B \times P_A$$

Example

- In $E_{97}(2,1)$ with $G=(17,1)$, Alice and Bob want to perform a Diffie-Hellman key exchange
 1. Alice picks $n_A = 13$ and Bob picks $n_B = 42$.
 2. Alice and Bob calculate
 1. Alice $P_A = n_A \times G = 13 \times (17,1) = (18, 90)$
 2. Bob $P_B = n_B \times G = 42 \times (17,1) = (91, 89)$
 3. They exchange P_A, P_B and:
Alicia calculates $K = n_A \times P_B = 13 \times (91, 89) = (29, 2)$
Bob calculates $K = n_B \times P_A = 42 \times (18, 90) = (29, 2)$

Cryptography with Elliptic Curves (cont.)

- It is also possible to encrypt with Elgamal's algorithm (which we do not see) and sign with DSA (coming soon)
1. **Efficiency** with a similar key size is similar to standard algorithms
 2. **Security** depends on the security of the logarithm in elliptic curves
 - Known attacks are slower than standard factorization and discrete logarithm attacks
- 1 y 2 $\Rightarrow$ To achieve equivalent security, we can use a much smaller key, and it's more efficient!

Comparative

Key sizes to achieve equivalent security levels:

Criptografía simétrica	RSA, DH, Elgamal	Curvas Elípticas
80	1024	160–223
112	2048	224–255
128	3072	256–383
192	7680	384–511
256	15,360	512 +

Recommended Curves

- Some elliptic curves are tested and optimized, and are recommended. Two examples:

Name	Values	Properties
Curve25519	$y^2 = x^3 + 486662x + 1$ $p = 2^{255} - 19$	- No ataques acústicos - Seguridad: 128 bits
Curve448-Goldilocks	$y^2 + x^2 = 1 - 39081x^2y^2$ <i>(hay una representación estándar más "fea")</i> $p = 2^{448} - 2^{224} - 1$	- Por su relación con el "número áureo" hay multiplicación eficiente - Seguridad: 224 bits

- Finding secure curves is a current topic of research in cryptography. For more information, please visit <http://safecurves.cr.yp.to>

Questions?

